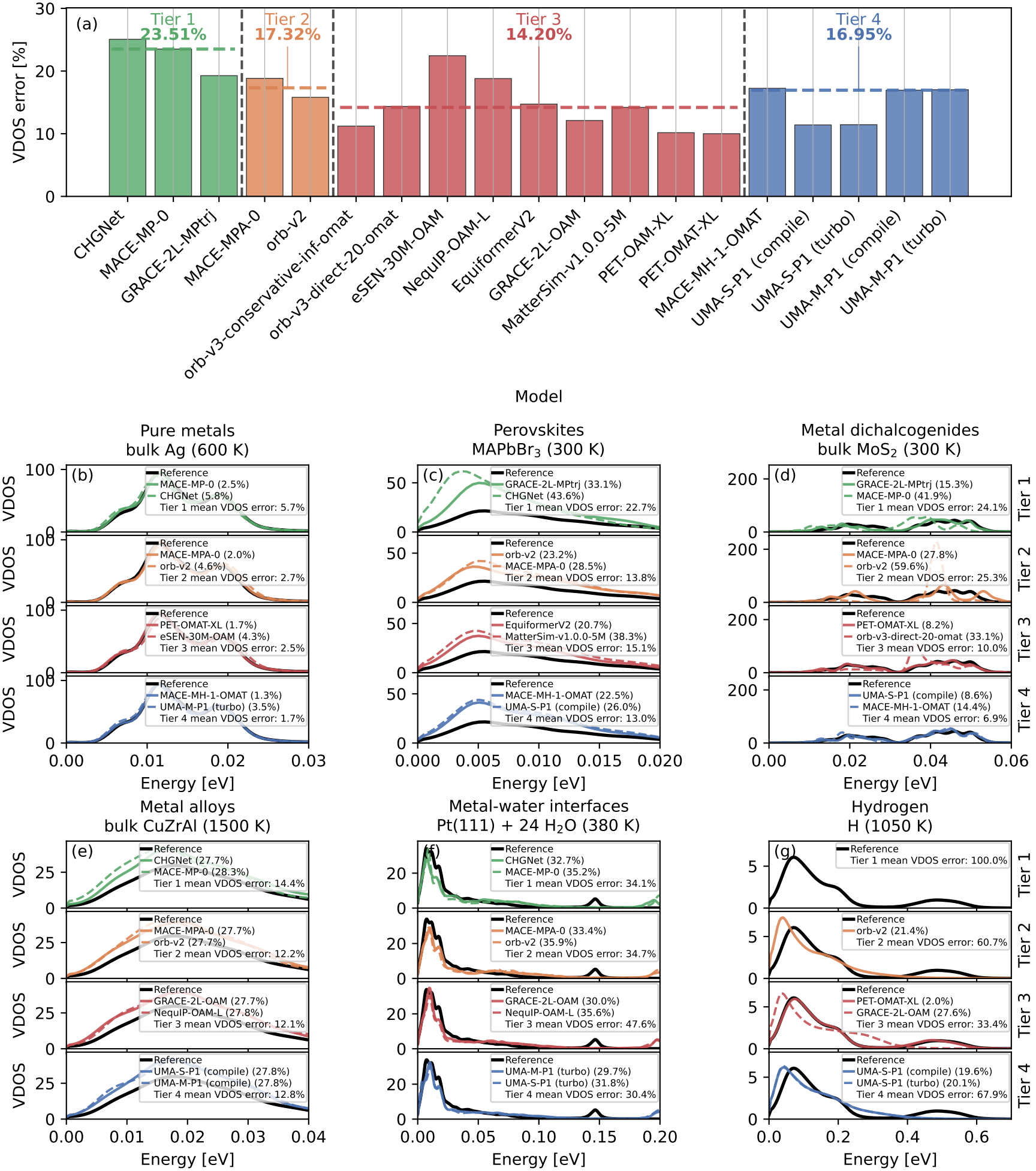

(a)
Tier 1
23.51%
Tier 2
17.32%
Tier 3
14.20%
Tier 4
16.95%
VDOS error [%]
CHGNet
MACE-MP-0
GRACE-2L-MPtrj
MACE-MPA-0
orb-v2
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
GRACE-2L-OAM
MatterSim-v1.0.0-5M
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
Model
Pure metals
bulk Ag (600 K)
(b)
Reference
MACE-MP-0 (2.5%)
CHGNet (5.8%)
Tier 1 mean VDOS error: 5.7%
MACE-MPA-0 (2.0%)
orb-v2 (4.6%)
Tier 2 mean VDOS error: 2.7%
PET-OMAT-XL (1.7%)
eSEN-30M-OAM (4.3%)
Tier 3 mean VDOS error: 2.5%
MACE-MH-1-OMAT (1.3%)
UMA-M-P1 (turbo) (3.5%)
Tier 4 mean VDOS error: 1.7%
VDOS
Energy [eV]
Perovskites
MAPbBr3 (300 K)
(c)
GRACE-2L-MPtrj (33.1%)
CHGNet (43.6%)
Tier 1 mean VDOS error: 22.7%
orb-v2 (23.2%)
MACE-MPA-0 (28.5%)
Tier 2 mean VDOS error: 13.8%
EquiformerV2 (20.7%)
MatterSim-v1.0.0-5M (38.3%)
Tier 3 mean VDOS error: 15.1%
MACE-MH-1-OMAT (22.5%)
UMA-S-P1 (compile) (26.0%)
Tier 4 mean VDOS error: 13.0%
Metal dichalcogenides
bulk MoS2 (300 K)
(d)
GRACE-2L-MPtrj (15.3%)
MACE-MP-0 (41.9%)
Tier 1 mean VDOS error: 24.1%
MACE-MPA-0 (27.8%)
orb-v2 (59.6%)
Tier 2 mean VDOS error: 25.3%
PET-OMAT-XL (8.2%)
orb-v3-direct-20-omat (33.1%)
Tier 3 mean VDOS error: 10.0%
UMA-S-P1 (compile) (8.6%)
MACE-MH-1-OMAT (14.4%)
Tier 4 mean VDOS error: 6.9%
Tier 1
Tier 2
Tier 3
Tier 4
Metal alloys
bulk CuZrAl (1500 K)
(e)
CHGNet (27.7%)
MACE-MP-0 (28.3%)
Tier 1 mean VDOS error: 14.4%
MACE-MPA-0 (27.7%)
orb-v2 (27.7%)
Tier 2 mean VDOS error: 12.2%
GRACE-2L-OAM (27.7%)
NequIP-OAM-L (27.8%)
Tier 3 mean VDOS error: 12.1%
UMA-S-P1 (compile) (27.8%)
UMA-M-P1 (compile) (27.8%)
Tier 4 mean VDOS error: 12.8%
Metal-water interfaces
Pt(111) + 24 H2O (380 K)
(f)
CHGNet (32.7%)
MACE-MP-0 (35.2%)
Tier 1 mean VDOS error: 34.1%
MACE-MPA-0 (33.4%)
orb-v2 (35.9%)
Tier 2 mean VDOS error: 34.7%
GRACE-2L-OAM (30.0%)
NequIP-OAM-L (35.6%)
Tier 3 mean VDOS error: 47.6%
UMA-M-P1 (turbo) (29.7%)
UMA-S-P1 (turbo) (31.8%)
Tier 4 mean VDOS error: 30.4%
Hydrogen
H (1050 K)
(g)
Tier 1 mean VDOS error: 100.0%
orb-v2 (21.4%)
Tier 2 mean VDOS error: 60.7%
PET-OMAT-XL (2.0%)
GRACE-2L-OAM (27.6%)
Tier 3 mean VDOS error: 33.4%
UMA-S-P1 (compile) (19.6%)
UMA-S-P1 (turbo) (20.1%)
Tier 4 mean VDOS error: 67.9%